

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       | Maths | Prac |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total |       |      |
| 7        | (a) | (i)   | Circuit sends current that can be varied through lamp (1)<br>Accept series variable resistor correctly drawn, variable potential divider, or any recognisable symbol for variable (dc) power supply.<br>Ammeter and voltmeter correctly connected and shown by correct symbols (1)                                                                                                                                                                                                                                                                                                                                                                                      | 2               |     |     | 2     |       | 2    |
|          |     | (ii)  | Choose 200 mA as better resolution (accept precision, accuracy)<br>Accept 0.16 A is closer to maximum of range for that meter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 | 1   |     | 1     |       | 1    |
|          |     | (iii) | More points needed [for lower voltages] (1) accept smaller intervals or more readings<br>pd range needs <u>extending</u> [beyond 2 V (to 3 V)] (1)<br>Don't accept take evenly spaced readings or take repeat readings or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |     | 2   | 2     |       | 2    |
|          | (b) | (i)   | 5.0 [ $\Omega$ ] <b>or</b> $\frac{0.50}{0.10}$ [ $\Omega$ ] <b>or</b> by implication (1)<br>13.8 [ $\Omega$ ] <b>or</b> $\frac{2.00}{0.145}$ [ $\Omega$ ] <b>or</b> by implication (1)<br>2.8 (1) Accept correct answer expressed as a fraction<br>14.2 [ $\Omega$ ] or 14.3 [ $\Omega$ ] are from graph mis-readings deduct 1 mark only                                                                                                                                                                                                                                                                                                                                |                 | 3   |     | 3     | 3     |      |
|          |     | (ii)  | <b>Marking points</b><br><b>M1</b> [Free] electrons make collisions<br><b>M2</b> .....with ions <b>or</b> atoms <b>or</b> lattice (accept particles)<br><b>M3</b> Increasing the pd increases the current / drift velocity / energy of electrons<br><b>M4</b> Electrons hit ions harder or transfer more energy [ <b>or</b> equivalent] when $I$ [or $V$ ] larger<br><b>M5</b> Increased vibration [amplitude] [of ions/atoms]<br><b>M6</b> Therefore temperature increases<br><b>M7</b> Collisions more frequent <b>or</b> more probable [when $T$ larger]<br><b>M8</b> Smaller current / drift velocity [for a given pd]<br><b>M9</b> Resistance increases in context |                 |     |     |       |       |      |